

Lokesh Goenka

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AI/ML Engineer specializing in Generative AI, conversational AI, and speech technologies. Experienced in building LLM-powered agents, RAG systems, LLM fine-tuning, and end-to-end voice AI pipelines. Published researcher with work spanning personalized learning and large language models.

SKILLS

Programming Languages: Python, SQL

Generative AI: LLMs, RAG, Agentic AI, Prompt Engineering, LLM Fine-tuning, LoRA, QLoRA

Speech & Voice AI: TTS, STT, Speech-to-Speech, Audio-to-Audio, Conversational AI, Voice Agents

NLP & ML: Transformers, Hugging Face, Embeddings, Tokenization, Scikit-learn, Model Evaluation

Frameworks & Libraries: PyTorch, FAISS, NumPy, Pandas, FastAPI, Google ADK, LiveKit

Cloud & Deployment: AWS, Docker, Git, OpenSearch, Vector Databases

Programming: Python, SQL

EXPERIENCE

Associate AI & ML Engineer

August 2025 – Present

Soulax, Hyderabad, Telangana, India

- Build real-time conversational AI agents using Google ADK, LiveKit, and AWS, with agent workflows for low-latency and context-aware voice interactions.
- Evaluate and integrate STT, TTS, and Audio-to-Audio models, focusing on latency, speech quality, model behavior, and production integration.
- Develop agent workflows for natural voice conversations, tool calling, contextual responses, and reliable multi-turn interactions.

AI & ML Intern

May 2025 – July 2025

Soulax, Hyderabad, Telangana, India

- Developed LLM-powered NL2SQL pipelines that convert natural-language questions into executable SQL queries.
- Contributed to RAG pipelines for grounded responses, improving factual accuracy and response quality.
- Researched reinforcement learning approaches for improving agent performance and user interactions.

Data Science Intern

June 2024 – August 2024

SIFY, Chennai, Tamil Nadu, India

- Contributed to a personalized learning platform for adaptive student learning experiences and automated assessment generation.
- Fine-tuned FLAN-T5 and Llama 3.1 models for automated multiple-choice question generation.
- Achieved 95% generation accuracy on educational assessment datasets.

Co-Founder

February 2023

Eco-Einsteins

- Spearheaded initiatives focused on **promoting environmental sustainability** and raising awareness about Eco-conscious practices within the student community.
- **Organized events and workshops** to educate and engage students on sustainable living and waste management practices.

PUBLICATIONS

- **Question Generation for Personalized Learning Platform for Students** (Springer Nature, 2026)
- **From Uniform to Unique: Adaptive K–12 Assessment Using Large Language Models** (LNCS, 2025)

PROJECTS

- Vedic Sanskrit TTS** | *Python, PyTorch, LLMs, Speech AI* 2026 – Present
- Researching and developing a TTS pipeline for Vedic Sanskrit mantra chanting with emphasis on pronunciation, Vedic svara, and high-fidelity speech generation.
 - Experimented with codec-language-model architectures, Indic-Mio/MioCodec, LoRA fine-tuning, codec tuning, and acoustic/token-level conditioning.
 - Built evaluation workflows for speech quality and investigated codec reconstruction, pitch/svara preservation, pronunciation, and long-form generation behavior.
- Personalized Learning Assistant for Students** | *Python, NLP* | [Github Repo](#) August 2024 – Present
- Developed an AI-powered Personalized Learning Assistant that identifies student strengths and weaknesses through adaptive assessments.
 - Fine-tuned Llama 3.1 using QLoRA for automated MCQ generation and evaluated generated questions against educational datasets.
 - Implemented Retrieval-Augmented Generation (RAG) using FAISS for educational content retrieval.
 - Designed adaptive assessment strategies using reinforcement learning for personalized question generation.
 - Published research findings in Springer Nature conferences.
- Saavaad** | *HTML5, CSS, React* | [Github Repo](#) January 2023 – March 2023
- Developed a website that helps in translating, summarizing text, and converts text to speech. It helps overcome language barriers, comprehend information efficiently, and enjoy content on the go.
- StorEtellor** | *HTML, CSS, JavaScript* | [Link](#) January 2021 – March 2021
- Developed a web platform enabling children to read interactive stories and learn valuable lessons.

EDUCATION

- Masters of Science, Data Science and Computing**
Sri Sathya Sai Institute of Higher Learning July 2023 – April 2025
- Bachelors of Science, Computer Science**
Sri Sathya Sai Institute of Higher Learning Aug 2020 – May 2023

CERTIFICATIONS

Data-driven Astronomy, [Coursera](#) | CS50X , edX | Graphic Design in Corel Draw , [Udemy](#)